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HOUSE OF ASPIRANTS

COMPUTER

CHAPTER 4

Input Devices

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CHAPTER 4

Input Devices

Computer Awareness

IN THIS CHAPTER

- Keyboards and mice
- Scanners and digitizers
- Joysticks and light pens
- OCR and MICR
- Touchscreens
- Biometrics

An Input Device is a hardware device that allows a user to enter data, instructions, or control signals into a computer.

The computer receives the input through input devices, processes it using the CPU (Central Processing Unit), and produces the desired output.

BASIC WORKING

User

Input Device

CPU (Processing)

Output Device

EXAMPLES OF INPUT DEVICES

- Keyboard
- Mouse
- Scanner
- Joystick
- Light Pen
- Digitizer
- Microphone
- MICR
- OCR
- Touch Screen
- Webcam
- Biometric Devices

1. KEYBOARD

A Keyboard is the most commonly used input device used to enter text, numbers, symbols, and commands into a computer.

Connectivity

- USB Keyboard (Wired)
- Bluetooth Keyboard (Wireless)

Types of Keyboards

A. QWERTY Keyboard

The most commonly used keyboard layout in the world.

The first six letters of the top row are:Q W E R T Y

Features

- Standard keyboard layout
- Usually contains 104 keys
- Widely used in India

Important Keys

Toggle Keys:Used to switch a function ON or OFF.

Examples:

- Caps Lock
- Num Lock
- Scroll Lock

Modifier (Combination) Keys:Used along with another key.

Examples:

- Shift
- Ctrl
- Alt

Exam Point

Most commonly used keyboard = QWERTY

B. AZERTY Keyboard

Used mainly in:

- France
- Belgium
- Some European countries

The first six letters are:A Z E R T Y

Exam Point

AZERTY = French Keyboard

C. DVORAK Keyboard

Dvorak was designed to improve typing efficiency; it does not guarantee that every user will type faster.

Exam Point

DVORAK = Alternative layout designed for typing efficiency

2. MOUSE

A Mouse is a hand-held pointing device used to control the cursor on the screen.

Inventor

Douglas Engelbart conceived the mouse in the early 1960s; Bill English built the first prototype in 1964.

Main Parts

- Left Button
- Right Button
- Scroll Wheel

Touchpad

A laptop uses a Touchpad instead of a traditional mouse.

The cursor moves when a finger is moved across the touchpad.

Types of Mouse

A. Trackball Mouse

Contains a ball fixed inside the device.The user rotates the ball using fingers.

B. Mechanical Mouse

Uses:

- Ball
- Rollers

Drawback

Collects dust and requires regular cleaning.

C. Optical Mouse

Uses:

- LED
- Optical Sensor
- Digital Signal Processing

Features

- No ball mechanism
- More accurate
- Faster

Important Fact

Microsoft popularized modern optical mice in 1999; optical-mouse technology existed earlier.

D. Wireless Mouse

Uses:

- Bluetooth
- Proprietary radio-frequency link (commonly 2.4 GHz)
- Infrared (IR)

Advantage

No cable required.

Important Term

Hovering

Moving the mouse pointer over an object without clicking it.

3. SCANNER

A Scanner converts printed documents, photographs, or images into digital form.

Uses

- Document scanning
- Image scanning
- Digital archiving

Types of Scanners

A. Flatbed Scanner

Most commonly used scanner.

Features

- Document placed on glass surface
- Scanning head moves

Uses

- Books
- Documents
- Photos

B. Handheld Scanner

Held in hand and moved over the document.

Example

Barcode Scanner

C. Sheetfed Scanner

Features

- Paper moves through scanner
- Scanner remains stationary

Best For

Single-page documents

D. Drum Scanner

Uses:PMT (Photomultiplier Tube)

Features

- Extremely high resolution
- Professional image scanning

E. Photo Scanner

Specially designed for photographs.

Features

- High color depth
- High resolution

4. JOYSTICK

A Joystick is a pointing device consisting of a stick mounted on a movable base.

Inventor:C. B. Mirick

Uses

- Video games
- Flight simulators
- Military applications

Difference from Mouse

Mouse	Joystick
Cursor moves only when mouse moves	Cursor keeps moving in stick direction
Used for general work	Used mainly for games

5. LIGHT PEN

A Light Pen is a pen-shaped pointing device.

It contains a light-sensitive sensor that detects objects on the screen.

Invented:Around 1955 Developed Under Whirlwind Project (MIT) Uses

- Drawing
- Selecting object

6. DIGITIZER (GRAPHICS TABLET)

A Digitizer is a flat input device used with a stylus.It converts hand-drawn images into digital form.

Uses

- Digital drawing
- Graphic design
- CAD (Computer Aided Design)
- AutoCAD
- Signature capture

7. MICROPHONE

A Microphone is an input device used to enter sound into a computer.

Functions

- Records voice
- Voice calls
- Video conferencing
- Online meetings

Examples

- Zoom Meetings
- Google Meet
- Online Classes

8. MICR(MAGNETIC INK CHARACTER RECOGNITION)

MICR reads characters printed with magnetic ink.

Mainly used in:

- Banks
- Cheque Processing

Exam Point

MICR is used on bank cheques.

9. OCR(OPTICAL CHARACTER RECOGNITION)

OCR converts printed or handwritten text into editable digital text.

Uses

- Digitizing books
- PDF conversion
- Document scanning

Example

Scanning a printed page and converting it into editable Word text.

OCR vs MICR

Feature	OCR	MICR
Full Form	Optical Character Recognition	Magnetic Ink Character Recognition
Technology	Optical	Magnetic
Used For	Documents	Cheques
Accuracy	High	Very High

10. TOUCH SCREEN

A Touch Screen acts as both:

- Input Device
- Output Device

Users interact by touching the display directly.

Uses

- Smartphones
- Tablets
- ATM Machines

- Kiosks

Frank Beck and Bent Stumpe developed a capacitive touchscreen at CERN in 1973; touchscreen designs existed earlier.

11. VIRTUAL REALITY (VR)

Virtual Reality (VR) creates a computer-generated environment that appears real to the user.

Devices Used

- VR Headset
- Gloves
- Headphones

Applications

- Gaming
- Training
- Medical Simulation
- Education

12. WEBCAM

Definition A Webcam is a camera connected to a computer.

Uses

- Video calls
- Online classes
- Recording videos
- Capturing images

Examples

- Laptop Camera
- USB Webcam

13. BIOMETRIC DEVICES

Biometric Devices identify a person using unique biological characteristics.

Examples

- Fingerprints
- Face
- Retina
- Iris
- Voice
- Palm Veins

Types of Biometric Devices

A. Face Scanner

Identifies a person using facial measurements.

Examples:

- Face Unlock in Smartphones

B. Hand Scanner

Uses palm vein patterns.

Feature

Palm-vein patterns are distinctive biometric characteristics; there is no universal ranking that makes them more unique than fingerprints.

C. Fingerprint Scanner

Most commonly used biometric device.

Uses

- Attendance Systems
- Mobile Phone Unlocking
- Security Systems

Exam Point

No two persons have identical fingerprints.

D. Retina/Iris Scanner

Uses the eye for identification.

E. Voice Scanner

Uses voice characteristics for identification.

COMPARISON TABLE OF IMPORTANT INPUT DEVICES

Device	Input Type
Keyboard	Text
Mouse	Pointing
Scanner	Image
Microphone	Sound
Webcam	Video
MICR	Magnetic Characters
OCR	Printed Text
Touch Screen	Touch
Fingerprint Scanner	Biometric

MEMORY TRICKS

Keyboard Types:

QAD

Q → QWERTY

A → AZERTY

D → DVORAK

Scanner Types

FHSDP

F → Flatbed

H → Handheld

S → Sheetfed

D → Drum

P → Photo

MOST IMPORTANT FACTS FOR EXAMS

- Douglas Engelbart invented the Mouse.
- QWERTY is the most common keyboard.
- AZERTY is the French keyboard.
- DVORAK improves typing speed.
- Barcode Scanner is a handheld scanner.
- MICR is used in banks.

- OCR converts printed text into editable text.
- Touch Screen is both input and output device.
- Light Pen is not used with LCD screens.
- Digitizer is used in CAD and AutoCAD.
- Retina Scanner provides high security.
- Fingerprint Scanner is the most common biometric device.

ONE-LINE REVISION POINTS

- Keyboard enters text.
- Mouse controls cursor.
- Scanner digitizes documents.
- Joystick is used in gaming.
- Light Pen is pen-shaped input device.
- Digitizer converts drawings into digital form.
- Microphone inputs sound.
- MICR reads magnetic ink.
- OCR converts images into text.
- Touch Screen is both input and output device.
- Webcam captures images and video.
- Biometrics identify individuals using body features.

QUICK REVISION

Keyboard → Text Input

Mouse → Pointing Device ,Inventor: Douglas Engelbart

Scanner → Image Input

Joystick → Gaming

Light Pen → Screen Selection Digitizer → Graphics & CAD

Microphone → Sound Input

MICR → Bank Cheques

OCR → Text Recognition

Touch Screen → Input + Output

Webcam → Video Input

Biometrics →

Face

Fingerprint

Palm

Retina

Iris

Voice